

HOW PREDICTIVE AI GOES WRONG

IN 2015, administrators at Mount St. Mary's University, a private university in Maryland, USA, wanted to increase their university's retention rate—the proportion of admitted students who go on to graduate. The school conducted a survey to find which students were struggling. This might sound like a worthy goal; once the administration knows who is struggling, they can offer additional support to those students to help them adjust to college. But instead, the president suggested dismissing students who weren't doing well. He reasoned that if these students dropped out in the first few weeks of the semester, as opposed to later, they wouldn't count as "enrolled" and therefore wouldn't decrease the university's retention rate.

In a faculty meeting, the president said: "My short-term goal is to have 20–25 people leave by the 25th [of September]. This one thing will boost our retention 4–5%."¹ Professors raised objections by pointing out that it is hard to judge who will be successful in the first few weeks after entering college. "This is hard for you because you think of the students as cuddly

bunnies, but you can't," the president responded. "You just have to drown the bunnies . . . put a Glock to their heads."

This is a startling example, but the fact is that many schools would like to predict which students are at risk of dropping out—some of them for reasons more aligned with students' well-being. An AI-based product called EAB Navigate could automate this process. In its marketing pitch to schools, EAB claimed: "The model will provide your school and its advisors with invaluable and otherwise unobtainable insight into your students' likelihood of academic success." Even if some schools might use this insight to pressure students to leave, others could conceivably use it to design interventions that might help students stay in school. But interventions that seem helpful could also be questionable. For example, the tool helps by recommending alternative majors in which a student would be more likely to succeed. This might have the effect of driving out poorer and Black students—whom the tool is more likely to flag—from more lucrative but more challenging STEM majors.² And throughout this process, students may have no idea that they are being evaluated using AI.

EAB Navigate is an example of an automated decision-making system that uses predictive AI. In this realm, there is a vast amount of AI snake oil.

Companies selling these tools make strong claims about their utility.³ They are deployed widely by governments and private companies. Still, unlike generative AI applications like ChatGPT, predictive AI has largely flown under the radar when it comes to public interrogation. In many cases, including EAB Navigate, the decision subjects don't even know they are being automatically evaluated.

In this chapter, we will see how predictive AI goes wrong. While a full account of these failures would take up more than

the entire length of this book,^{4,5*} we will highlight common failures that are nevertheless hard to fix, underscoring the difficulty of making predictive AI work.

Before we dive in, let's look more closely at how these automated decisions are made.

Predictive AI Makes Life-Altering Decisions

Algorithms like EAB Navigate are everywhere, and they are used to automate consequential decisions about you—with or without your knowledge.⁶ When you show up at a hospital, an algorithm could determine if you need to be admitted overnight or released the same day. When you apply for public benefits such as child welfare, an algorithm could determine if your application is valid or if you are trying to commit fraud. When you apply for a job, an algorithm could determine if your application will be considered by a recruiter or discarded. When you visit a beach, an algorithm could determine if the water is safe to swim in.

An algorithm is a set of steps or rules used to make a decision. Sometimes these rules are developed by people or institutions. For example, during the COVID-19 pandemic, the U.S. government sent stimulus checks to help citizens deal with the economic hardships they faced. Adults were paid USD 1,200; children were paid USD 500. Policymakers decided these rules. But once the rules were established, people's eligibility was automatically determined based on their past tax records.

* Two popular incident databases record failures of AI, including predictive AI, in the real world. The AI Incident Database has over 600 reports, and the AI, Algorithmic, and Automation Incidents and Controversies Repository has over 1,400 reports, as of 2024.

1. If the applicant is a U.S. citizen and eighteen or older, send a check for USD 1,200.
2. If they are a U.S. citizen and under eighteen: send a check for USD 500.
3. If they're not a U.S. citizen or earn more than USD 75,000 per year*: don't send a check.

This type of algorithm, in which the rules are developed manually but applied automatically, is commonly employed in the public sector, such as in allocating funds for public housing or welfare.

But increasingly, algorithms are used to *develop* the rules automatically from patterns in past data. For instance, while watching Netflix, if you rated *Forrest Gump* highly but not *The Shining*, a recommendation algorithm might predict that you'll enjoy drama films over horror films. Here, a Netflix employee is not manually creating a rule stating that users who liked *Forrest Gump* should be recommended other drama films. Instead, your ratings and the time you spend watching different types of films determine what films you're recommended next. Unlike the first kind of algorithm, the rules are both developed and applied automatically.

This kind of algorithm is called a model, which is a term you might encounter frequently. A model is usually a set of numbers that mathematically specify how the system should behave. These numbers may not be intelligible to a human, even to the developer of the system, unless the model is explicitly designed with that goal in mind. Models are created from data, or "trained," using a set of statistical techniques called machine learning.

* The actual algorithm was slightly more complex: the amount paid if a person earned more than USD 75,000 tapered off. And there were different rules for people with children.

These models can be much more consequential than deciding your next Friday night movie. They are used for allocating scarce resources, such as jobs or loans. They provide certain opportunities to people and foreclose other opportunities. This is what we call predictive AI: models used for decision-making based on predictions about the future, such as who will do well at a job or who will pay back a loan.

Take the example of criminal justice, where predictive AI is used for many kinds of decisions: Should an inmate get parole? What should happen to people who are arrested? Before trial, a judge needs to decide whether a defendant should be detained in jail, released on bail (and if so, what the bail amount should be), or released without any bail, but perhaps with other restrictions such as ankle monitors.

A lot rides on the answer.⁷ Spending time in jail can ruin someone's life. They can lose their income in the short term—and even if they are released, especially if they have restrictions such as ankle monitors, it will still be much harder to find a job. They suffer from increased physical and mental illnesses, both due to stigmatization and because of poor conditions in jail. And a large fraction of the people in jail are simply there because they can't afford high bail amounts.

As a result, the criminal justice system disproportionately burdens the poor and leads to cycles of poverty and racial inequality. Almost half a million people are in U.S. jails at any given time without having been convicted of a crime.⁸ Despite violent crime in the country going down by almost 50 percent, the number of people jailed in the last four decades has nearly doubled.^{9,10}

Many U.S. states mandate the use of risk assessment tools to decide whether a person should be released or detained before trial, and predictive AI is often used. The tools produce two

main risk scores. One is the risk that the defendant will commit a crime, especially a violent crime, if released. The other is the risk that they will fail to appear in court on the appointed date. In each case, defendants are labeled low, medium, or high risk. The tools try to calculate these scores based on certain features of the defendants.

Let's zoom in on one predictive AI product: COMPAS, which stands for Correctional Offender Management Profiling for Alternative Sanctions. COMPAS relies on defendants' answers to 137 questions to make decisions about them.¹¹ It includes questions about their past history of crime or failure to appear for court cases. It also includes questions about things individuals have little or no control over, such as how often their family members have been arrested, or whether their friends or acquaintances have been arrested. And some questions seem to punish people for their personality or for poverty, such as "How often do you feel bored?" and "How often do you have barely enough money to get by?"

The developers of COMPAS aim to predict if a defendant will fail to appear in court or be arrested for a crime within two years.¹² It is trained using data on defendants' behavior in the past. COMPAS uses this past data to find patterns in the characteristics of defendants who failed to appear in court, such as their age, number of past offenses, and the criminal history of their peers, and tries to distinguish them from those who did appear for their trial. This highlights an assumption built into much of predictive AI: people with similar characteristics will behave similarly in the future.

Predictive AI is quickly gaining in popularity. Hospitals, employers, insurance providers, and many other types of organizations use it. A major selling point is that it allows them to reuse existing datasets that have already been collected for

other purposes, such as for bureaucratic reasons and record keeping, to make automated decisions.

Yet, predicting the future is hard. People could face unexpected setbacks such as getting evicted, or surprises such as winning a lottery, that no model can predict. Small changes in someone's life, such as a visit to the emergency room, could have large effects on their future—say, due to compounding medical bills.

Claims about the virtues of predictive AI are pervasive. Upstart's model decides whether an applicant should be approved for a loan.¹³ The company claims that its model is significantly more accurate than traditional lending models. It also claims to be a leader in fair lending practices and promises that future versions of the model will continue to be fair. Finally, it claims to be highly efficient: three-quarters of its loan decisions require no input from humans. Other companies make similar promises. HireVue sells tools to automate hiring decisions. It claims to predict how well a candidate will perform at their job once hired. HireVue's home page announces its selling points: "Fast. Fair. Flexible. Finally, hiring technology that works how you want it to."

Despite these statements, human decisions still come in at many stages of the development of predictive AI, often hidden from sight. And since the data used to train predictive AI is itself created using human discretion, we cannot guarantee that the decisions will be unbiased or fair.¹⁴ In other words, decisions made using predictive AI may still be very human.

We were skeptical of the promises made by developers of predictive AI, and so we decided to dig deeper. Together with researchers Angelina Wang and Solon Barocas, we spent over a year reading hundreds of research papers, news articles, and

reports about automated decision-making systems. We were surprised to find that the same set of shortcomings plague many applications of predictive AI.¹⁵ In the next few sections, we will take a closer look at these shortcomings, through real-world case studies.

Spoiler: We think predictive AI falls far short of the claims made by its developers.

A Good Prediction Is Not a Good Decision

Healthcare workers make a critical decision when a patient comes to the hospital with symptoms of pneumonia: whether to send the patient home after treatment or admit them overnight. They consider a patient's age as well as preexisting medical conditions, such as asthma, which place them at high risk if they get pneumonia. High-risk patients with pneumonia symptoms are typically sent straight to the ICU to avoid complications.¹⁶

In a 1997 study, researchers investigated whether AI could make better decisions than healthcare workers in predicting outcomes of patients with pneumonia.¹⁷ Like countless other AI researchers, they thought a model trained with a sufficiently large amount of data would make better decisions than a human, decisions which would help to prioritize high-risk patients.

The researchers trained an AI model and found that it had fairly good accuracy in predicting which pneumonia patients would face complications or death. Surprisingly, the model found that having asthma leads to *lower* risk of complications due to pneumonia. If this model were to be used in a hospital, it would be more likely to send an asthmatic patient home than a non-asthmatic one (let alone sending asthmatic patients to the ICU). How is this possible?

The researchers looked more closely. They found that in the dataset, asthmatic patients were indeed at a lower risk for serious pneumonia or death—but this was only because the training data was collected under the hospital’s preexisting decision-making system.¹⁵ Asthmatic patients were sent straight to the ICU as soon as they arrived, thereby receiving more intensive care than non-asthmatic patients and thus becoming less likely to develop complications.

So the model’s predictions were correct—but they were correct under the hospital’s existing system. Ironically, the model was meant to replace this system.

Deploying this model would have led to disastrous decisions. Asthmatic patients would have been perceived as lower risk by the model and sent home. Thankfully, the researchers realized what had happened and did not use the model in this particular hospital.

This error shines a light on a fundamental limitation of much of predictive AI: AI can make good predictions *if nothing else changes*. But correlation is not causation (and having asthma does not reduce the risk of complications from pneumonia). In other words, predictive AI does not account for the impact of its own decisions. That is, it can’t tell us what would happen if something changed in a system—if, for instance, the model started sending asthma patients home.

Let’s look at another example from healthcare. A 2018 study claimed to accurately predict hypertension using machine learning, with impressive results.¹⁸ On closer look, however, it became clear that the model was evaluated on people who had already been seen by clinicians.¹⁹ This meant that a critical input to the model was whether a patient already used drugs for controlling hypertension. But if a patient is using antihypertensive drugs, they have obviously already been diagnosed with hyper-

tension. These cases were still counted as successes when evaluating the model, vastly inflating its accuracy numbers.

This issue arose in part because the researchers relied on existing data rather than new data collected specifically for the task. Generally, collecting data is expensive and time-consuming. So, some AI developers claim that existing data is enough to make good decisions.

The importance of collecting new data is widely recognized in medicine, which relies on randomized controlled trials (RCTs) to test the impact of introducing a new drug or vaccine. To test if a drug performs well, half the people in the trial are randomly assigned to the experimental group and are given the drug. The other half are assigned to the “control” group and are given a placebo. The groups are carefully balanced with respect to demographics like age and gender. A drug’s effectiveness is evaluated based on whether it leads to lower disease rates in the experimental group compared to the control. Medical researchers perform RCTs despite their slow pace and high expense for a simple reason—easier, faster methods don’t work. The same is true in many areas where AI is used for automated decision-making.

Unfortunately, predictive AI companies don’t seem to have realized the importance of collecting data on the impact of their tools’ decisions. And such data collection is also expensive and time-consuming, so it would negate their claims of cutting costs and increasing efficiency.

As a result, even if AI can make accurate *predictions* based on past data, we can’t know how good the resulting *decisions* will be before AI is deployed on a new dataset or in a new setting. So, when you encounter claims about the effectiveness of predictive AI, it’s important to find out if developers evaluated the impact of the decisions or only the accuracy on past data.

Opaque AI Incentivizes Gaming

We've seen that we can't fully determine the impact of AI before deployment, in part because AI makes its predictions assuming that the system will remain the same as it was during testing. But systems change, and people are one obvious wild card. Things become even more complicated when people behave strategically.

The colonial British government in India wanted to reduce the cobra population, so it decided to offer a reward to people who brought in dead cobras. But instead of killing cobras in the wild, people started breeding them to claim the reward—leading to an *increase* in the cobra population. This is an apocryphal example of a mismatch between what is specified and what is desirable. A similar problem occurs in AI. When developers build AI, they specify the exact outcome they want to predict. But because of the reliance on past data, they are forced to choose this outcome based on the data that is already available. As a result, what AI predicts can be a poor proxy for what we want it to predict.

A stark example is AI used for hiring. In the United States, three-fourths of employers use automated tools to screen job candidates.²⁰ Automated hiring tools come in various forms. Some are used to filter candidates based on their résumés. Others test candidates using an automated video interview. Yet others require them to solve puzzles. These tools are used as the first step in selecting candidates. If someone doesn't pass, their résumé is discarded without a human ever looking at it. But the process is opaque; companies don't share how their software is built, and candidates are left in the dark about what criteria they are judged on.²¹

In response, candidates have developed strategies to work around opaque hiring AI. They stuff their résumés with keywords

from the job application and add the names of top universities in white text (which a human reader can't see, but a computer can recognize).²² In video interviews that they know will be judged by AI, they try using fancy words such as "conglomerate."²³

It is not clear whether these strategies work. So a group of journalists decided to investigate. They looked at Retorio, a Munich-based startup that offers hiring tools based on video interviews.²⁴ Their findings were surprising: simple changes to a person's appearance, such as wearing a scarf or glasses, were enough to drastically change the score that the AI tool produced. Adding a bookshelf or a painting in the background increased scores, while making a video darker (without changing its content) led to lower scores. In another study, researchers looked at personality tests used in hiring and found that merely changing the format of candidates' résumé from PDF to plain text changed their personality scores.²⁵

Changing the background or format of a résumé obviously does not change someone's capability to perform well at a job. So why do candidates' scores change? One reason could be that in the data used to train the model, people who had bookshelves in the background performed better compared to those who had a plain background.

This is precisely the kind of correlation that candidates banked on when they added keywords to their résumé. They assumed people with those keywords in their résumé performed better in the past. They wanted to avoid getting rejected just for omitting the right keywords or not using enough fancy words.

Acting strategically—gaming—to avoid negative consequences is common. This is what teachers do when they teach to the test, and what consumers do when they try to increase

their credit scores without changing their spending habits, such as by getting a retail credit card or filling out a prequalification form before applying for credit. But with AI-based hiring, people don't even know what actions influence their chances. Instead of guiding candidates toward decisions that actually improve their skills, AI-based hiring encourages candidates to experiment with superficial changes to their résumé and application.

We are not taking a stance on whether candidates should or shouldn't be attempting these tricks. The ethics of gaming are interesting but tangential to our point. Rather, our point is that when AI companies make claims about accuracy, they fail to account for the effects of people behaving strategically. When the model's output can be easily manipulated using superficial changes, we cannot take claims of accuracy at face value. Opaque models also impose a cost on decision subjects in terms of wasted time, such as when candidates experiment with ways to add invisible fake qualifications to their résumés.

Overautomation

In 2013, the Netherlands deployed an algorithm to flag welfare fraud, replacing an earlier system in which humans reviewed each decision.²⁶ The algorithm was used to make serious accusations of guilt using only statistical correlations in data, without any other evidence.²⁷

This shift to an automated system had many adverse effects. For one, people lost the ability to challenge decisions. Inaccurate or outdated government data often leads to erroneous fraud allegations, which under the new system could not be reversed. Further, the data used to make these allegations was not publicly available, so people had no way to find out why they'd been accused of fraud.

Over the next few years, the algorithm wrongly accused around thirty thousand parents of welfare fraud in childcare payouts. The amounts the government said they owed were in some cases over a hundred thousand euros—sending many parents into mental and financial ruin.²⁸ Shockingly, the algorithm used nationality to predict if someone had committed fraud. All else being equal, people were more likely to be flagged for fraud if they had Turkish, Moroccan, or Eastern European nationality.²⁹

Despite the algorithm's shortcomings, the country used it for six years. When details of the algorithm were shared in 2019, the public was outraged. The Dutch data protection watchdog investigated privacy failures caused by the algorithm's use. It fined the country's tax authorities—who had built the algorithm—EUR 3.7 million. This was the largest such fine in the country. In 2021, the prime minister and his entire cabinet resigned, in part due to the use of the welfare fraud algorithm.

This is an example of overautomation: when AI is used to make decisions without offering any recourse to decision subjects. Overautomation has led to other notable failures in fraud detection—even when AI is not in use. Using an algorithm to detect unemployment fraud, the U.S. state of Michigan wrongly collected USD 21 million from residents between 2013 and 2015.³⁰ And the Australian government incorrectly collected AUD 721 million from its citizens between 2016 and 2020 in what's been called the Robodebt scandal.³¹

To protect themselves from accusations of overautomation, developers of these systems often include fine print saying that they should always be used with human supervision. But this merely passes the buck and doesn't necessarily have the intended effect.

In the summer of 2022, Toronto used AI to predict when people should avoid swimming at public beaches due to high bacteria levels.³² The developer claimed the software achieved over 90 percent accuracy in predicting water safety. But it did much worse: on 64 percent of the days when the water was unsafe, beaches remained open based on incorrect assessments.

When journalists questioned city officials about the efficacy of the predictive tool, officials responded by saying it wasn't used on its own—a human overseer always made the final decision. Journalists later found that the supposed overseers *never* changed the decisions that the software spit out.

This is a familiar pattern. AI developers use a bait and switch when it comes to human oversight. They sell predictive AI based on the promise of full automation. Eliminating jobs and saving money is a big part of their pitch. But when AI fails, developers retreat to the fine print, saying that it shouldn't be used without human oversight.

Even if that oversight technically exists, it is often inadequate due to limitations of time, expertise, or authority. The bureaucrats in charge may be overworked, or may not have the training to challenge the automated decisions or the incentive to stick their necks out to do so.

In one extreme case, U.S. health insurance company UnitedHealth forced employees to agree with AI decisions even when the decisions were incorrect, under the threat of being fired if they disagreed with the AI too many times. It was later found that over 90 percent of the decisions made by AI were incorrect.³³

Even without such organizational failure, overreliance on automated decisions (also known as “automation bias”) is pervasive. It affects people across industries, from airplane pilots to doctors. In a simulation, when airline pilots received an

incorrect engine failure warning from an automated system, 75 percent of them followed the advice and shut down the wrong engine. In contrast, only 25 percent of pilots using a paper checklist made the same mistake. If pilots can do this when their own lives are at stake, so can bureaucrats.³⁴

No matter the cause, the end result is the same: consequential decisions about people's lives are made using AI, and there is little or no recourse for flawed decisions.

Predictions about the Wrong People

AI reflects its training data. It learns patterns about the people who make up the data, and the decisions made by AI reflect these patterns. But when the decision subjects come from a population with different characteristics than those in the training data, the model's decisions are likely to be wrong. For instance, a predictive AI system that performs well in one country could be utterly useless in another.

Let's look at how this plays out in two criminal risk prediction systems in the United States: the Ohio Risk Assessment System (ORAS) and the Public Safety Assessment (PSA). Similar to COMPAS, both attempt to predict the risk of releasing a defendant before trial.

ORAS was trained on data from 452 defendants from the U.S. state of Ohio in 2010 and is now used across the United States. There are a few obvious problems with this. Ohio's patterns of criminal activity might differ from those in other states. The data doesn't represent everyone. The small number of people who were used to create the model could otherwise vary from the larger population it was eventually used on. Finally, as with any model, accuracy could degrade as patterns of crime change over time.³⁵

By contrast, PSA was trained using 1.5 million people from three hundred jurisdictions across the United States. It is used in over twenty states. At first glance, this seems to solve part of the problem. If the model is trained on a much larger dataset from many jurisdictions, it should be more accurate when applied to the entire country. But nationwide trends of criminal activity can differ from local trends.

Cook County in Illinois is a stark example. It adopted PSA in 2015. But the rate of violent crimes there is much less than the national average. Compared to the training data, *ten times* fewer defendants who were marked as “high risk” went on to commit violent crimes. PSA used data from all over the country, but it did not consider that crime rates could be much lower in some counties. Thousands of defendants were unnecessarily jailed for months before trial—based only on a model’s prediction, without any evidence of their guilt.³⁶

PSA erred because it didn’t distinguish data from different counties—it made predictions about the wrong people. In some cases, getting data about the entire population of interest is impossible, making this error hard to resolve.

One example comes from Allegheny County, Pennsylvania. In 2016, the county adopted the Allegheny Family Screening Tool to predict which children were at risk of maltreatment.³⁷ The tool is used to decide which families should be investigated by social workers. Through these investigations, social workers can forcibly remove children from their families and place them in foster care.

The tool relies on public welfare data, which consists mainly of data on poorer parents who use public services such as Medicaid-funded clinics. Notably, it doesn’t include information about people who use private insurance.³⁸ Models built using this data therefore cannot make decisions about rich par-

ents who have never relied on public services. As a result, the tool disproportionately targets poorer families.

This is an example of how AI tools search under the streetlight. More often than not, the streetlight is pointed at the poor.

Whenever predictive AI is deployed, it is critical to ask: Who was it tested on? When predictive AI is built using one population but is used on another, claims about how well it performs are based on insufficient evidence.

Predictive AI Exacerbates Existing Inequalities

The cost of flawed AI is not borne equally by all. The use of predictive AI disproportionately harms groups that have been systematically excluded and disadvantaged in the past.

A notable example comes from predicting who should receive better medical care. With the passage of the Affordable Care Act in the United States in 2010, insurance companies started asking hospitals to provide services at lower prices, threatening to remove providers who did not comply. One of the main ways hospitals cut costs was by identifying high-risk patients and providing them with preemptive care so as to avoid future expenses, such as hospitalization. To identify high-risk patients, they turned to AI.

Dozens of models were developed to assess patients' health risks. Developers claimed that AI could rank patients based on their healthcare needs and assign additional resources to the ones classified as high risk. While the pneumonia risk prediction model that we saw in the beginning of this chapter was never deployed, health risk prediction tools used to reduce hospital expenses are deployed across the United States.

Enrollment in a high-risk healthcare program has huge consequences. It dictates whether a patient will receive preemptive

care and personal assistance. Still, most developers did not share how they built their models—until recently, we knew little about how well their products performed.

One such product is Optum's Impact Pro. In 2019, researchers studied it and found that based on its predictions, Black people were less likely to be admitted to high-risk programs than White people. In other words, a Black person who had the same health risks as a White person would receive poorer care.³⁹

On closer inspection, the researchers found the culprit. It is hard to quantify how much a patient needs healthcare. But hospitals know exactly how much people *spend* on healthcare. So instead of predicting healthcare *needs*, Optum chose to predict how much an insurance company will *spend* on healthcare.

Higher healthcare costs don't always reflect higher health needs. Higher bills could occur due to access to better insurance; more time, attention, and care at the doctor's office; or more visits to the doctor. Or perhaps those with higher bills are simply those who can afford to pay higher costs for co-pays and deductibles. There are clear inequalities in the U.S. healthcare system, and the use of Optum's predictive AI would mean that people who were already receiving better healthcare would be more likely to be classified as high risk and continue to receive better care in the future.

Unsurprisingly, in the training data used for creating the model, Black people received poorer care than White people, even those with similar health conditions. The tool had a racial bias because the predicted quantity (healthcare costs) was a poor stand-in for what the developers claimed to predict (healthcare needs or a patient's risk level).

Optum's use of healthcare costs to measure risk made sense for its business. Its clients were hospitals that wanted to reduce

costs. Even after this damning study was published, the company continued using healthcare costs to build its model.

Business incentives are one of many reasons predictive AI increases inequality. Another is developers' reliance on past data. As we have seen, collecting new data for developing predictive AI is expensive and time-consuming. But existing data might not have information about what developers want to predict (in this case, patients' healthcare needs or their risk levels). So, developers use proxies that are easier to measure and already exist in the data (such as healthcare costs).

Let's look back at COMPAS, the tool which provides a pre-trial risk score. Its developers claim that judges can use this score to predict whether a defendant will commit a crime or fail to show up for their trial. But the data used by COMPAS does not contain information about crimes. It only includes information on who was arrested. The difference is subtle but salient. Not all crimes lead to arrests—in many cases, they can be undetected or ignored. Police could arrest someone only to later find them innocent in court. And there are well-known racial disparities in policing—Black people in the United States are more likely to be arrested than White people for the same crimes. These differences can create chasms between what companies claim to predict (crime) and what they actually predict (arrests).

For all these reasons, when predictive AI systems are deployed, the first people they harm are often minorities and those already in poverty. We've already seen other examples, such as Allegheny County's child maltreatment prediction tool that only had data about those who access public benefits such as child welfare. And we'll see many more in the chapters to come.

A World without Prediction

Why is predictive logic so pervasive in our world? We think a major reason is our deep discomfort with randomness. Many experiments in psychology show that we see patterns where none exist, and we even think we have control over things that are, in fact, random.^{40,41} When people are forced to confront randomness and the illusion of control breaks down, they look for ways to take back control.

A good example is election forecasting. In the United States, presidential elections occur every four years, and forecasting the winner is a common practice, starting well over a year in advance. But it is far from an exact science. On the morning after the 1948 presidential election, the *Chicago Tribune* infamously blared “Dewey Defeats Truman” on its front page.⁴² Of course, Truman had won the election and would be the next U.S. president. In a hurry to call the election before the paper went to press, the newspaper relied on poll outcomes to preemptively predict the (incorrect) winning candidate. And seventy-five years later, things are only slightly better. The 2016 election forecasts were famously incorrect, predicting Hillary Clinton defeating Donald Trump.

Despite its limitations, forecasting has become a sort of spectator sport, and people’s fascination with it can border on obsession. Here’s a quote from a *Wired* article about one such person:

When he wakes up in the morning, he doesn’t shower or eat breakfast before checking the Nate Silver-founded site’s presi-

* Harry S. Truman and Thomas E. Dewey were the Democratic and Republican presidential candidates in the 1948 elections.

*dential election forecast. He keeps a tab open to FiveThirtyEight's latest poll list; a new poll means new odds in the forecast. He gets push alerts on his phone when the forecast changes. He follows the 538 Forecast Bot, a Twitter account that tweets every time the forecast changes. In all, Evan says he checks in hourly, at least while he's awake.*⁴³

This is despite poll numbers not affecting most people's daily lives in any meaningful way. Why does this happen? It all comes down to our inability to handle the uncertainty hanging over our lives. Uncertainty avoidance goes one level deeper than being averse to risk. Even if the election forecaster says the chance of our favorite candidate winning is 50-50, that somehow feels better than not knowing at all.

What does this have to do with AI snake oil? Here's our thesis. It's true that companies and governments have many misguided commercial or bureaucratic reasons for deploying faulty predictive AI. But part of the reason surely is that decision-makers are people—people who dread randomness like everyone else. This means they can't stand the thought of the alternative to this way of decision-making—that is, acknowledging that the future cannot be predicted. They would have to accept that they have no control over, say, picking good job performers, and that it's not possible to do better than a process that is mostly random.

And what if a decision-maker did embrace unpredictability? Suppose a company announced that it would hire candidates randomly after filtering out clearly unqualified candidates, and it would promote employees randomly among those who met performance criteria. Because the overemphasis on merit is so ingrained in so many people, the company would come to be seen as an undesirable place to work, and it would lose

good candidates. Such a policy wouldn't survive long in our world.

In fact, when such lotteries were used for allocating housing, people participating in those lotteries had a largely unfavorable opinion of them.⁴⁴

And so, the discomfort we experience with randomness can lead to a search for patterns where none exist. This paves the way for biases. Consider the business mantra, "No one ever got fired for buying IBM." Opting for the familiar, the tried-and-true, is safer than embracing uncertainty. Even the practice of hiring primarily from elite colleges reflects a desire to impose order on the inherently unpredictable process of identifying talent and potential.

Accepting the inherent randomness and uncertainty in many of these outcomes could lead to better decisions, and ultimately, better institutions. Instead of treating people as fixed and their outcomes as predetermined, we need to work on building institutions that are genuinely open to the fact that the past doesn't predict the future. Such a world is possible—if only we can learn to embrace the randomness that underpins our lives. We'll return to this theme in the final chapter.

Concluding Thoughts

As predictive AI is deployed, it is important to understand how it fails and how it harms people, in order to avoid falling prey to snake oil. Change starts by challenging the deployment of harmful AI tools in your workplace, neighborhood, and community. An informed public is the first step in demanding change.

In this chapter we encountered many reasons why predictive AI fails. Table 2.1 recaps them. But given the increasing amount

TABLE 2.1. Five reasons predictive AI fails

Reason	Example
A good prediction can result in a bad decision.	Patients with asthma could be sent back home when they come to a hospital with symptoms of pneumonia.
People can strategically game opaque AI.	Adding bookshelves in the background increases scores on automated hiring tools.
Users over-rely on AI without adequate oversight or recourse.	The Dutch welfare fraud detection model falsely accused 30,000 parents of fraud without any recourse.
Data for training AI may come from a different population than the one it is used on.	PSA's criminal risk prediction relied on a national sample. It overestimated the risk in counties where crime was rarer.
Predictive AI can increase inequality.	Optum's Impact Pro led to an increase in the difference in the quality of care between Black and White patients.

of data collected about people, as well as advances in machine learning, it might seem like the limitations we have seen are temporary. On the other hand, it is also possible that no matter how much data we have or how good our models become, there are inherent limits to how predictable the future is. Which of these scenarios is more likely? The next chapter will answer this question.

WHY CAN'T AI PREDICT THE FUTURE?

ARE OUR LIVES PREDETERMINED? Can we predict what will happen to us tomorrow? Next year? In our careers? And if it is possible to predict our futures, how do we go about doing it? These questions are not new. Humans have always wanted to know the future. Historically they've relied on the predictions of individuals who claimed to be "gifted" in discerning the future. Kings consulted oracles before they went to war. People went to fortune tellers in an unpredictable world where diseases could strike at any moment.

Rather than relying solely on the purported clairvoyance of certain individuals, people throughout history also developed prediction systems that anyone could conceivably master.¹ These systems were structured ways of predicting the future. They were based on the positions of the stars, or tarot cards, or lines on the palm, or dozens of other things.

Today, the prediction system of choice is AI. Many people believe that we can use algorithms and machine learning to analyze current and historical facts in order to make accurate predictions about future events. An apocryphal example of

gaining surprising insights from data is an unlikely story of beer and diaper sales.² When analyzing sales data from a chain of stores, a data scientist learned that the sales of beer and diapers were correlated—presumably because new parents don't have time to go to a bar for a drink, but they can drink at home. So, if stores stocked these two items next to each other, the sales of both items would increase because when parents were shopping for diapers, they would also pick up beer (and vice versa).

This type of insight seems like the perfect example of what we can understand from data. After all, there could be millions of such correlations possible in a typical store, and it is believable that AI could indeed tell us which of these correlations we should take seriously. Similarly, techno-optimists have argued that statistical tools can replace the scientific method because of the sheer number of correlations they can find in data that no human scientist could conceive of.³

It's understandable why this is a popular idea—why we might believe that computers, through sheer number crunching, can foretell what's to come. Unlike other methods of predicting the future, the power of computers doesn't require a suspension of disbelief—after all, computing, and AI specifically, has worked extremely well in many areas, such as transcribing speech or creating images from text. So, to the average person, it doesn't feel like too much of a stretch that a company could develop a tool to predict future outcomes, say, one that tells police where and when crime will occur.

At the same time, in the last chapter, we saw that predictive AI often doesn't work as claimed, and it can cause a lot of harm. Are these limits inherent to predictive AI? Or will current limits be overcome as AI methods advance and we gain access to more data?

Given the rapid uptake of predictive AI, in 2020, Arvind co-taught a course with Matt Salganik at Princeton University that collated evidence on how well we can predict the future.⁴ This chapter is based on the lessons from that course and the research Sayash and Arvind have done since then. The main questions that we will address are: How well can AI predict individual and societal outcomes, and where does it fall short? In which cases can we hope to improve our predictions about the future, and in which cases are limits to predictability inherent?

Let's begin by looking at the computing tools we use to predict the future.

A Brief History of Predicting the Future Using Computers

When your weather app tells you there's an 80 percent chance of rain, you're witnessing the end result of a millennia-long quest to understand the skies. In Egypt, before the construction of the Aswan High Dam, the Nile flooded every year. Starting in the third century BC, and continuing for about a thousand years, priests observed the rising level of the Nile to predict water levels for each farming season.⁵ If the level reached a stone marker by a certain date, priests would know the flooding that year would be adequate for farming.

Fast-forward a thousand years. By the end of the seventeenth century, several meteorological instruments had been created: hygrometers to measure humidity, thermometers to measure temperature, and barometers to measure pressure. In the nineteenth century, weather-observing stations were set up across the globe. Compiling data from different locations around the world could enable meteorologists to predict, to some extent, how the weather would evolve. In the early twentieth century,

equations to simulate weather patterns were proposed, but it wasn't until the 1940s that the computational power required for these calculations caught up. John von Neumann, one of the most prominent mathematicians in the world at the time, directed the construction of a computer at the Institute of Advanced Study in Princeton (less than two miles from where we are writing this book). Among other applications, it could calculate weather prediction estimates using the simulation equations that had already been proposed. By the mid-1950s, regular weather forecasts were being made across North America.⁶

But a 1963 finding raised doubts about our ability to predict the weather well—especially for longer time horizons.⁷ Meteorologist Edward Lorenz was trying to predict weather patterns using simulation equations. During one of these calculations, he made a chance discovery: rounding the numbers to three decimal places instead of six gave vastly different results!⁸

This finding led to a profound scientific advance—the recognition that weather is a chaotic system. That is, small changes in initial conditions, like a small error in measuring temperature, lead to exponentially increasing errors later. The farther away the prediction, the larger the error. Lorenz termed this the Butterfly Effect: the flapping of a butterfly's wings in Brazil can cause a tornado in Texas.⁹ What this means is that at least in principle, a butterfly flapping its wings could have ripple effects on the atmosphere that grow larger and larger with time. Once scientists understood this effect, predicting the weather over longer time periods started to seem like a herculean task.

But something interesting has happened in the last few decades. Increased computational power, more data, and better equations for simulating the weather have led to weather

forecasting accuracy increasing by roughly one day per decade. A five-day weather forecast a decade ago is about as accurate as a six-day weather forecast today.¹⁰ Of course, it remains impossible to predict the weather a year out or even a month out. To do that, you'd really need to know the position of every butterfly. But a one-week forecast is still extremely valuable. This improvement is not the result of a revolution in computational methods, theory, or data collection capabilities. Rather, it is the end result of consistent small improvements.

Weather prediction today largely relies on a computational tool known as simulation. The idea is that the future evolution of a system can be predicted using two crucial pieces of information: the current state and equations describing how the system changes over time based on the interactions between its components.

Given the successes of predicting the evolution of physical systems, one might hypothesize that with the right data and enough computing power, simulations could be used to predict any type of event. In the 1950s, MIT professor Jay Forrester pioneered an approach called system dynamics, which tried to extend the use of simulations to social systems.¹¹ He tried to use simulation to model a whole city, with an eye to solving thorny social problems like urban blight.

Here, things went badly wrong. And we don't have to speculate too much to find out why—a city is far from a physical system. A city can't be perfectly observed, even in a totalitarian society, so there is a lot of uncertainty in the starting conditions of a simulation. Further, a city is not a self-contained system, so unless you're simulating the whole world, your simulation will go off track. In fact, Forrester's models didn't account for suburbanization at all—the growth of the suburbs as residential areas, with people commuting into cities for work. This was

perhaps the most significant shift that American cities were undergoing at the time of his work.

In the '60s, a company called Simulmatics—a portmanteau of *simulation* and *automatic*—aimed to predict outcomes like U.S. elections, the effectiveness of counterinsurgency efforts in Vietnam, and race riots.¹² As far as we can tell, it didn't actually use simulation at all, just basic statistics and algebra.¹³ Still, the company's choice of its name was perhaps indicative of the extent to which people at the time associated simulation with prediction. But Simulmatics, too, was largely a failure, and the company shut down in 1970.

In contrast to simulation, machine learning uses past data to learn underlying patterns and make predictions about future events, often in ways that can adapt and change over time. There are no fixed rules about how the future will play out given past events. Rather, these rules are determined based on how the system behaved in the past.

For example, weather simulations typically rely on physical laws about the world, such as equations describing airflow over time. But if we were to use machine learning, we would use data to build computer models of how the weather has evolved in the past, and then use those models to predict how it will evolve in the future. We would no longer need to rely on an understanding of physics. Nor would we need to predict the weather minute by minute or hour by hour, with each prediction being the input to the next. We would instead directly model the relationship between the weather today and the weather tomorrow, or whatever time horizon we're interested in.

Broadly speaking, machine learning is more suited for predicting things about individuals whereas simulation is more suited for predicting collective or global outcomes. That's because there are millions of people about whom you can

collect training data to build machine learning models. For example, spam classifiers today work reasonably well, because there are lots of examples of both spam and nonspam emails.

But if you're predicting whether there will be a food shortage, machine-learning-based predictions are not that useful, because there aren't enough examples of food shortages in the past that can be used to train a machine learning model. Here, simulation based on domain knowledge, such as how changing climate conditions affect agriculture, the intricacies of global trade, and the impact of political shifts on food supply can be more useful.

Around the same time as the fall of Simulmatics, the seeds for more sophisticated machine learning methods were taking root. FICO* was established in 1956 to predict individuals' credit risk. The company started by creating models for individual lenders and banks. Much later, in 1989, the company launched the general-purpose FICO score, which could be used as a measure of creditworthiness across financial institutions. The FICO score used the credit history of individuals to predict how likely they would be to default on a loan in the future.¹⁴ Similarly, in the 1980s, criminal risk prediction became widespread.¹⁵ Soon after the turn of the twenty-first century, machine learning became commercially successful in serving personalized ads, and then in many other areas. It started to be seen in the tech and business worlds as the default approach to almost any decision problem. Following this success, the use of predictive AI exploded in the 2010s. The vast majority of large companies started using some form of hiring automation.¹⁶ And we'll see many more examples soon.

* Then known as Fair, Isaac, and Company.

Despite the widespread use of computational predictions in social settings, you might be surprised to learn that there's remarkably little published evidence of their effectiveness. So, how well can we really predict future social outcomes? Let's dive in.

Getting Specific

Not all predictions are hard. Some phenomena can be predicted with a high level of accuracy. Astronomers can predict planetary movements and the life cycles of stars. Doctors can predict the progression of some diseases—and they can even predict how different drugs affect that progression. Given these successes, what are phenomena that we *can't* predict?

Based on what we saw in the previous chapter, a first guess could be that we can't predict any social phenomena. But even this is too broad. We can certainly predict some social phenomena reasonably well, like the amount of traffic on a route, or how busy a store would be on a certain day.^{17,18} So, clearly, we need to be more specific.

One area where predictions seem hard is people's futures. Can we predict a student's GPA a year in advance? What about predicting if someone will be evicted from their current home? In the next section, we'll review evidence of how accurate such predictions can be. But first, let's pause to ask how accurate a prediction needs to be in order to be considered good or bad. This question might seem simple, but it turns out to be surprisingly nontrivial.

For virtually any kind of prediction, there are many different ways to judge its accuracy, ways which may produce results that seem drastically different from each other. When trying to predict the weather, do we consider a prediction accurate if the

temperature is within one degree? Or is it enough to predict it within five degrees? Or do we only care if we can correctly predict if it will rain, regardless of temperature? Any of these criteria can help us compare two different methods for prediction or determine whether our ability to predict the weather is improving over time. But there isn't one true way to judge if a weather prediction is good or bad.

This also means we can't directly compare how good a prediction in one area is to a prediction in another area. That is, we can't make statements like "weather patterns are harder to predict than sales patterns." It isn't meaningful to ask whether predicting the temperature within one degree is better or worse than predicting the sales amount in a store within 10 percent.

Still, there are many *qualitative* criteria that can help us understand whether prediction tasks can be done well. Weather forecasting isn't perfect, but it can be done well enough that many people look at the forecast in their city every morning to decide whether they need an umbrella. But we can't predict if any one person will be involved in a traffic accident on their way to work, so people don't consult an accident forecast every morning.

This comparison highlights another important quality of predictions: we only care about how good a prediction is in relation to what can be done using that prediction. For example, our ability to predict earthquakes is excellent—if the goal is to identify high-risk zones that need to have stringent building codes.¹⁹ But if the goal is to identify when an evacuation needs to take place because of an imminent earthquake, our prediction ability is basically nonexistent.²⁰ The first task is about where earthquakes might strike, while the second task is about when.

In other cases, such as in criminal risk assessment, prediction could be practically useful from the perspective of those in

power—say a court system trying to manage jail populations—yet morally dubious in deciding who should be jailed and who should be released.

A third, related criterion is whether the accuracy of the prediction improves with more data and better models. Suppose a disease is the result of a complex but deterministic genetic process. That is, it depends completely on someone's genes. Then, with enough genetic sequencing and improved models, we can hope to predict whom it will affect with high accuracy. On the other hand, suppose a type of cancer is the result of radiation exposure leading to a random genetic mutation. We might be able to say that someone has an elevated risk because they are exposed to radiation in their job, but beyond that risk assessment, we can't predict if they will develop cancer, no matter how much data we collect, because of the randomness inherent to this process.

For both diseases, the judgment of high and low predictability is not based on absolute numbers. In fact, in numerical terms, given our current (relatively poor) knowledge of genetics, genetic disease may be far less predictable than cancer. Yet the two trajectories of predictability differ as our data and scientific knowledge increase. Genetic diseases become more predictable over time, whereas there are fundamental limits to the predictability of cancer, just like we can't predict the outcome of the roll of a fair dice, no matter how much data we collect or how good our AI is.

So when we say life outcomes are hard to predict, we are using a combination of these three criteria: real-world utility, moral legitimacy, and irreducible error, that is, error that won't go away with more data and better computational methods.

Let's return to the question of why life outcomes are hard to predict. One of the most obvious explanations is that people

have agency. This explanation is unsatisfying to us. Of course, people have agency, but how often do we exercise it? Maybe we just don't exercise it often enough, and as a result we are still largely predictable. Besides, even if people aren't inherently predictable in a state of nature, the kinds of things that organizations want to predict, like students' GPAs, are constructs that are specifically designed and chosen for the purpose of exerting control over individuals. So, they may be much more predictable than other aspects of people's behavior.²¹

To arrive at better answers, let's look at empirical evidence from the real world.

The Fragile Families Challenge

Predicting the future is a focus in many scientific fields, but not in social science. Instead of predictions, the dominant approach in social science is to focus on improving our *understanding* of what causes phenomena of interest.²² For example, sociologists generally do not aim to predict individuals' future income for targeted interventions. Rather, they aim to understand the causes of poverty so that interventions can be targeted toward alleviating poverty.

Similarly, while social science helps us understand the causes of incompatibility and divorce in marriages, it isn't used to predict whether a couple will file for divorce.²³ In the 1990s and early 2000s, sociologists did attempt this, and they generated a lot of excitement about predicting which couples would divorce. But these studies were plagued with faulty methods that led to vastly exaggerated claims of predictive accuracy.²⁴ Essentially, the models researchers used would perform well for people whose information was included in the dataset used to train the model, but not on couples whose information wasn't

in the dataset. In other words, prediction only worked well in cases where the answer was already known!

One reason for the focus on understanding over prediction is the lack of available data. Until recently, sociological datasets were small. Machine learning works best when applied to large datasets, so sociologists have historically relied on simple statistical models like linear regression.

As the amount of available data has increased, the use of machine learning for prediction has made inroads in the social sciences. Let's take a look at one such attempt, the Fragile Families Challenge, a notable large-scale study that tried to predict children's outcomes using AI and lots of data.

In 2015, our colleague at Princeton University, Matt Salganik, wanted to study how well AI could predict the future. Sara McLanahan, also a Princeton sociology professor at the time, had been following the lives of over four thousand children who were born around the year 2000 in over twenty U.S. cities. Over the previous fifteen years, she and her colleagues had surveyed each child and their family—at birth, and at the ages of one, three, five, and nine. Through these surveys Sara and colleagues collected over ten thousand data points about each child using detailed surveys from parents, teachers, and in-home activities with the child. In fact, you'd have trouble finding a sociological variable of interest that *wasn't* included in the surveys.

In 2015, Sara and colleagues were planning to release their latest round of survey data, collected when the children reached the age of fifteen. Matt wondered if it was possible to use survey data from the Fragile Families project to see how well AI could predict the future. He walked down to Sara's office to discuss the details, and the seed of a collaboration was planted.

Matt and Sara felt that a single research group analyzing this data wouldn't suffice to evaluate whether children's life outcomes

are predictable. It might give a lower bound on accuracy, but no matter how good the results were, someone might object that a different model could perform better. Or that a different group of better-skilled researchers would build better models. In other words, it is hard to prove a negative result in this setting.

To address this concern, Matt and Sara organized a prediction competition, together with Princeton colleagues Ian Lundberg and Alex Kindel. A subset of the data—all data from birth until age nine—was released to competition participants across the world. Using this data, participants were asked to create AI models that would predict how well the children were doing at age fifteen. They would make predictions about six outcomes for children, such as their GPA, whether they had been evicted, and whether their household was facing material hardship. Participants would be ranked based on how close they were to the true outcomes of the children.

Since the competition was open to the public (and hundreds of researchers participated), different teams could try out different methods. Some used complex AI models. Others used sociologically informed statistical models. Regardless of their approach, researchers would compete on an equal footing: the only thing that mattered was how well their model predicted future outcomes of the children. The goal wasn't to pick a winner, but to learn from the collective effort. In fact, the organizers called it a "mass collaboration."

But unlike a regular collaboration, the prediction competition would help avoid overoptimism about how well AI could work. Many AI studies exaggerate how well a model can predict the future, because the researchers building the model have access to the data used to evaluate it. This means they can overestimate how well the models would do in the real world (and this is exactly what happened in the divorce prediction studies we

mentioned earlier). In contrast, the Fragile Families organizers didn't release the data used to evaluate the models, so none of the teams could game the results before submitting their predictions, even unintentionally.

Eventually, 160 teams submitted their predictions. Among the models being compared was a simple model relying on basic statistical techniques, which was intended to serve as a baseline for comparing more complex AI models. This baseline model consisted of just four features: three related to the mother of the child and one related to the outcome at age nine. For example, for predicting GPA at age fifteen, the model consisted of the mother's race, marital status, and educational level, and how well the student did in school at age nine.

To Matt's surprise (and disappointment), none of the models performed very well—the best models were only slightly better than a coin flip. And complex AI models showed no substantial improvement compared to the baseline model consisting of just four features.²⁵

In other words, tens of thousands of data points about thousands of families, a hundred and sixty competing researchers, and the most advanced AI models were not much better at predicting the future than decades-old regression models that used well-established sociological theories. The data showed that past GPA, race, and social class are useful in predicting future GPA. Sociologists had long understood these trends, so they were not exactly headline news.

Why Did the Fragile Families Challenge End in Disappointment?

Before Matt presents the results from the Fragile Families Challenge in academic talks, he usually asks the audience to guess how the teams performed. Some of the most optimistic audience

members are computer scientists and data scientists, who have seen how well AI can perform in other domains.

When confronted with the disappointing result, this is also the group with the most follow-up questions and ideas for improving predictions. One of the most common questions: Are samples from four thousand families enough? They often refer to another contest that set off the deep learning revolution in 2012. It was a contest known as the ImageNet Challenge that asked participants to identify the contents of images using AI, and it consisted of 1.2 *million* labeled images.²⁶ (We'll get back to ImageNet in more detail in the next chapter.)

One possibility for improving social predictions is exactly the sort of brute-force intervention that computer scientists suggested in this case—use samples from more people. The hypothesis is that greater amounts of data and more computational power would lead to drastically better accuracy and in turn a breakthrough in social prediction.

This is one of many reasons we can't conclude that the results of the Fragile Families Challenge indicate fundamental limits to prediction. We just do not know whether this hypothesis is correct. In scientific fields whose theories are more or less settled, such as planetary orbits in astronomy, there is high predictability. We can predict a planet's position years from now with astounding accuracy. In other cases, there are fundamental limits to predictability, and we know what those limits are. For instance, the laws of thermodynamics help us estimate the behavior of a gas like oxygen or nitrogen as a whole, but not the behavior of individual gas molecules.

But so far, we do not have theories about the predictability of social problems. We can't predict the future well, *and* we don't know what the fundamental limitations of our predictions are.

The predictability of life outcomes has been explored over and over in science fiction. The premise of the dystopian science fiction film and short story *Minority Report* is that would-be criminals can be apprehended based on predictions about crimes that they have not yet committed. The plot of *Person of Interest*, a TV drama, centers around AI that can predict crimes. The fundamental tension in these pieces is usually between pre-determination and free will, yet they ignore one important source of irreducible error in predicting social outcomes: chance events.

One notable feature of tasks where AI seems to work well, such as classifying what's in an image, is that once you have an image, say an image of a cat, it is easy to say what's in it. In this case, the irreducible error is small; humans (and AI these days) can correctly classify most images. Chance plays little role in determining the correct answer.

How high is this irreducible error for social predictions? Our understanding of social science and our theories of predictability are not yet at the point where we have a concrete answer, but there are reasons to think that error is high, in part due to chance events. People can face shocks that are completely unpredictable and that have a large effect on their life trajectory. No model can predict if someone might win a lottery or get hit by a car, for instance.

But how often do these unpredictable events happen? It might be true that a butterfly could cause a tornado, but if this happens once every millennium, maybe we don't need to worry about it. Much more common than large shocks are small initial advantages that are compounded over time. For example, a small bias in annual performance reviews (say because you have an antagonistic boss) could lead to large effects over the course of your career (you may not be promoted as quickly as others).

The difficulty of measuring these small differences leads to higher irreducible errors in predictions.

Let's come back to how much more data we would need to predict future outcomes well. We know that the number of samples needed to create accurate models increases sharply as the samples become more noisy. And social datasets have a lot of noise. In addition, the patterns underlying social phenomena are not fixed. Unlike images of cats, they differ greatly based on different contexts, times, and locations. What identifies success at one place and time might be completely uninformative for predicting success at another.

This means that we might need a lot of data from different social contexts for AI to predict the future well. Simply using past data is not enough—just like it's not enough to use polling data from a previous election to predict if a Democrat or Republican would win the next U.S. presidential election.

This raises an interesting possibility. Perhaps collecting enough data to make accurate social predictions about people is not just impractical—it's impossible. Matt Salganik calls this the eight billion problem: What if we can't make accurate predictions because there aren't enough people on Earth to learn all the patterns that exist?²⁷

Just as important as the number of samples is what's in those samples. In the Fragile Families Challenge, each child had around ten thousand sociologically relevant features recorded in the data. But you could imagine this, too, not being enough. Here's why.

After the prediction competition concluded, Matt and colleagues tried to find out why these models had performed so poorly.²⁸ So they decided to meet the families who had been the most unpredictable, to find out what caused the predictions about them to be so incorrect. In one interview, they found that

a struggling child had suddenly improved their performance in school. The reason? A supportive neighbor who had counseled the child, provided help with homework, and fed them blueberries. But there was no question in the Fragile Families data that asked children if someone outside the family was feeding them blueberries (or, perhaps more impactfully, helping them with homework). Could this be an example of a missing feature that, if present in the data, would lead to accurate predictions—say, because it signifies the presence of a supportive adult in the child's life? How many such features are today's datasets missing?

One way to build a more expansive dataset that could be used for more accurate predictions is to rely on data collected by governments. For example, the Netherlands has compiled data about people's families, neighbors, classmates, households, and colleagues. This dataset is extensive; it has records on each person living in the country—a total of 17.2 million people. On average, each person is connected to eighty-two other people, for a total of 1.4 *billion* network relationship records.²⁹

Such data is obviously much larger and more complete than the Fragile Families Challenge dataset. It is a realistic alternative that *could* be used to predict social outcomes. If it is indeed possible to use this data to predict future outcomes of interest, we'll find out soon enough. There are several efforts, including a prediction competition, that are currently underway to test this hypothesis.³⁰

Another possibility for additional data collection is via tech companies. People already spend significant amounts of time on platforms owned by companies like Google and Meta. Could the data they provide to these companies have insights that wouldn't otherwise be available?

We can indeed speculate about this possibility, as have many pop culture explorations of the relationship between technology

and society. But at the end of the day, making predictions about people's life outcomes is too risky, reputationally and legally, for tech companies. And in any case, their business metrics are not tied to predictions about people's future outcomes, but rather what content people will engage with today. So, it is unlikely we'll settle the question of the long-term predictive power of online data any time soon.

An even more grand (and dystopian) possibility is to collect extensive information on each person everywhere, to create überdatabases about people. In a world like this, everyone would be under constant surveillance so their every action is constantly monitored. The NSA and Big Tech companies already have a trove of data about people, but we're talking about something far more invasive—tracking every spoken word, every movement, every action, and perhaps even every electrical signal in their brain.³¹ Would such a world enable better predictions? To what end? And at what cost to our privacy?

Predictions in Criminal Justice

Ultimately, there aren't many examples of social science research on predicting life outcomes. And the evidence we do have suggests that we're quite bad at it. But as we saw in the last chapter, that hasn't stopped companies and governments from deploying tools to predict the future and make decisions about people based on those predictions.

There are dozens of AI tools and products made by companies that claim to predict individuals' futures.³² These predictions are used to make life-changing decisions about people in healthcare, insurance, banking, and criminal justice.

Through these deployments, we can try to assess how predictable life outcomes can be. While companies have resisted

giving out information about how their predictive AI works, researchers, journalists, and advocates have still managed to finagle information using companies' sales materials, interviews with employees, and public records requests. So we can look at how predictive AI has fared in the real world, and analyze the limits to prediction in these domains.

For example, one selling point of tools like COMPAS is that they can reduce bias and increase accuracy in the criminal justice system.³³ Since AI doesn't have the same biases that humans do, the story goes, it could be used to make decision-making objective and fair.

In 2016, Julia Angwin and her colleagues at ProPublica set out to investigate these claims. They ran an in-depth study on the use of COMPAS in Florida's Broward County to assess how predictions about ten thousand people panned out.

What they found was a big racial disparity. Among people who would *not* go on to commit crimes, Black people were twice as likely to be misclassified as high risk compared to White people.³⁴ Specifically, 45 percent of the Black people who didn't go on to commit crimes were marked as high risk, compared to just 23 percent of White people.

This racial bias became the subject of many academic studies and press reporting on risk assessment tools. But the report also revealed that the tool wasn't very accurate to begin with; it had a relative accuracy of just 64 percent.³⁵ Relative accuracy is calculated by looking at pairs of people where one would go on to commit a crime and the other wouldn't, and calculating how often the former was ranked as higher risk than the latter. An accuracy of 50 percent can be achieved by generating risk scores randomly! So a questionnaire that uses 137 data points about a person is only slightly better than deciding a defendant's fate based on a coin flip.

From a defendant's perspective, even if they have no intention of committing a crime, they could be jailed simply because their answers were similar to those of others who had gone on to commit a crime in the past.

Even the abysmal accuracy of COMPAS is likely to be an overestimate. To some extent, COMPAS is just predicting the biases of policing. For example, in a hypothetical world where everyone has the same probability of committing a crime in the future, but some people are more likely to be arrested (say, because they live in overpoliced areas), COMPAS would still be good at predicting future outcomes, just because people's locations determine their likelihood of being arrested. We have evidence of such biases in policing today, so there's little reason to expect that COMPAS isn't just predicting these biases.³⁶

Follow-up studies had results that were even more damning than ProPublica's original findings. One study showed that COMPAS was no more accurate than the judgments of nonexperts.³⁷ In fact, instead of using 137 data points, using just two data points about an individual—their age and their number of prior offenses—was as accurate as COMPAS. The lower the age and the higher the number of prior offenses, the more likely someone is to be rearrested.

There's cause for skepticism even in this simple rule. According to the data, younger defendants are riskier, but from a moral perspective, we might want to treat younger defendants more leniently because they are still developing neurologically, are more susceptible to peer pressure, and are more capable of long-term change.³⁸ If we take age out of the equation, we are left with just one feature: the number of past offenses.

So a simple rule—the higher the number of past offenses, the higher the risk—might be more morally legitimate than one that uses 137 data points about a person. And it has other positive

side effects: the rule is easily understood, people can challenge incorrect decisions, and it doesn't transfer power to unaccountable third parties (unlike COMPAS, a black box tool developed by a for-profit company).

Failure Is Hard. What about Success?

To recap, the last few sections have shown that in many settings it is hard to predict who will experience negative outcomes. Perhaps this is because failure is less about people's intrinsic qualities and more about circumstance. Surely, predicting success is different? Isn't it merit that determines who is successful? Being the fruit of years of hard work, success must be more predictable than, say, a spur-of-the-moment crime.

Well, the evidence suggests that if anything, luck plays an even bigger role in success than in failure.

Let's start with a personal story. Today, Arvind has a successful academic career. He applied to nine prestigious graduate programs back in 2004. Then the rejections started pouring in. After he was rejected by eight schools, he emailed the ninth school to ask if they had any news. They said they had no record of his application.

He panicked, then reached out to a friend who was a graduate student at the school in question. His friend learned that the application had landed in some other pile by accident. Arvind's disbelief was surpassed only by his amazement when the school actually accepted him. If it weren't for the friend willing to locate his application, his graduate school aspirations would have ended in a wipeout.

Getting into graduate school was only one step. Looking back at his career, each rung of the academic ladder involved a dose of luck.

All successful people are lottery winners to some extent. But they rarely acknowledge the role of luck in their success. Arvind isn't an exception—he only started telling his graduate school story publicly when he started writing about the role of luck, when it helped him make his point. This tendency might be especially true in the United States, because the view of society as meritocratic is so prevalent. As E. B. White said, “Luck is not something you can mention in the presence of self-made men.”³⁹ It's understandably hard for successful people to admit that they aren't necessarily deserving of their successes and it's all too easy for them to fool themselves into thinking that it was the result of talent and hard work (after all, talent and hard work do factor into success to some extent).

In sports, books, and films, success is hard to predict.⁴⁰ Tom Brady, widely regarded as one of the greatest quarterbacks in NFL history, was selected 199th in the sixth round of the 2000 NFL draft, which means almost every team passed on him multiple times. His big break came when Drew Bledsoe, the Patriots' starting quarterback, was injured.⁴¹

Harry Potter was rejected by eight publishers, John Grisham's debut novel by twenty-six, and Dr. Seuss's first book by twenty-seven. Orwell's *Animal Farm* was rejected because “it is impossible to sell animal stories in the U.S.” Perhaps hardest to believe is a book from the 1950s, which publishers rejected with comments such as “very dull,” “a dreary record of typical family bickering,” and “even if the work had come to light five years ago, when the subject [World War II] was timely, I don't see that there would have been a chance for it.” The book was *The Diary of a Young Girl* by Anne Frank.

Turning to films, *Star Wars* was rejected by United Artists and Universal Pictures. 20th Century Fox picked it up, but didn't put much faith in it, paying George Lucas only \$200,000

to write and direct it (the film went on to make \$461 million). On the flip side, while Disney is known for creating fan favorites, the 2012 sci-fi *John Carter* lost over \$100 million, and the 2022 animated feature *Strange World* lost around \$150 million.^{42,43} Studio executive David Picker said, “If I had said yes to all the projects I turned down, and no to all the other ones I took, it would have worked out about the same.”

So what’s going on here? There’s an underlying scientific reason it’s hard to predict success in each of these domains. If you believe in the idea of a meritocratic society, this is going to be a bit depressing.

Keep in mind that there are far more cultural products—books, movies, music—being produced than any one person can possibly consume in a lifetime. And only a tiny fraction of these can be bestsellers or blockbusters.

But why should there be blockbusters at all? Why does the success of books and movies vary by orders of magnitude? Are some products really thousands of times “better” than others? Of course not. A big chunk of the content that is produced is good enough that the majority of people would enjoy consuming it.

The reason we don’t have a more equitable distribution of consumption becomes obvious when we think about what such a world would look like. Each book would have only a few readers, and each song only a few listeners. We wouldn’t be able to talk about books or movies or music with our friends, because any two people would have hardly anything in common in terms of what they’ve read or watched. Cultural products wouldn’t contribute to *culture* in this hypothetical scenario, as culture relies on shared experiences. No one wants to live in that world.

This is just another way to say that the market for cultural products has rich-get-richer dynamics built into it, also called

“cumulative advantage.” Regardless of what we may tell ourselves, most of us are strongly influenced by what others around us are reading or watching, so success breeds success.

The depressing part is this: among the vast universe of “good enough” cultural products, it is a largely random process that determines success.⁴⁴ This is a mathematical consequence of cumulative advantage. The effect of an initial review of a book or rainy weather on the opening weekend of a film can get amplified over time. A noted actor signing on might attract other famous actors, leading to success-breeds-success dynamics during the film production process.

Studios and music labels hate this unpredictability. They have taken many steps to try to limit the effects of unpredictability, such as spending heavily on advertising and relying on franchises and sequels. But there is only so much they can do to resist these built-in dynamics.

A neat experiment to test this theory was conducted many years ago by a team led by our colleague Matt Salganik, who also led the Fragile Families Challenge. The researchers created a music app and recruited over fourteen thousand participants to rate and download songs from unknown bands. They wanted to understand the relationship between the quality of music and its popularity. To measure perceived quality, they asked a sample of participants to rate and download songs without being told how others had rated them or how often they had been downloaded.

The clever part was that the rest of the participants were put in one of eight different “worlds,” or different copies of the app. Each of these had the same songs but different consumption patterns (such as downloads) independent of each other. Each world listed the download counts within that world when a user logged on, which allowed the researchers to see how people

responded to this signal of their peers' assessment of different songs.

There was a lot of randomness in the results. Many mediocre songs did really well and many good songs did poorly. The same song could do very well in one "world" and poorly in another. In each world, songs that did well initially tended to continue to do well due to cumulative advantage. And in the world where people *couldn't* see download counts, there was much less inequality between songs that did well and those that didn't, evidence for the rich-gets-richer effect of showing download counts.

Let's return to career outcomes. Why does success hinge so heavily on luck? The unpredictability of career success is harder to study scientifically, because you can't put people into different "worlds" and experiment on them. The next best option is a "natural experiment," in which the natural course of events mimics the condition of an experiment. For example, suppose people early in their career apply for an important opportunity. Their applications are scored, and those above a cutoff receive the opportunity and those below it do not. However, there is no essential difference between the applicants immediately above and below the cutoff—say between those who score 65.1 percent versus 64.9 percent, and the cutoff is then decided at 65 percent. Which side of the line they landed on is essentially the luck of the draw. If we then follow those people's careers over time, we can experimentally test the effects of that initial success or rejection.

That's exactly what happened in the Netherlands, where grant proposals by recent PhDs are scored, and funding is awarded to those who score high enough. When researchers compared those immediately above and below the cutoff, they found that over the next eight years, the first group accumulated

twice as much funding as the second, even though, on average, the two groups were equally deserving.⁴⁵ This is evidence for a massive rich-get-richer effect, which can be hard to predict in advance because of the role of luck in determining who gets the first grant.

The Meme Lottery

The social media equivalent of a blockbuster or a bestseller is the viral hit; the main difference is that a social media post's success or failure is determined on an accelerated timescale compared to a book or movie. A tiny fraction of videos or tweets go viral while the rest get little engagement. One study found that fewer than one in one hundred thousand tweets had over one thousand retweets.⁴⁶ It's not just a matter of some users being popular. Even for the same user, popularity tends to be highly variable from one piece of content to another. On YouTube, an account's most viewed video is forty times more popular than its median video. On TikTok, it's sixty-four times more popular.⁴⁷ It is that sliver of viral content that dominates our attention.

This inequality is not necessarily engineered by the platforms, although platforms do try to amplify it. It naturally emerges from word-of-mouth dynamics.

In the early days of social media, this came as a surprise. One of the first viral videos on YouTube was "Charlie Bit My Finger."⁴⁸ It's a video of two boys, a three-year-old and a one-year-old, having a cute moment together. The baby innocently bites his brother, causing some pain, but it's all good by the end. The boys' father put it on YouTube to share with friends, but once there, it exploded. Within a couple of years, it was YouTube's most viewed video, and it would eventually receive almost a billion views.

At first, this confused many commentators. Why did this video go viral? There were many explanations, such as the fact that the older child goes through a wide range of emotions within a few seconds, making the video rewarding to watch. That seems plausible, but this and all other explanations were offered with the benefit of hindsight. We don't think the video's success could have been predicted in advance. In fact, when the father made the video, he didn't even consider it worth sharing online, posting it a few weeks later seemingly as an afterthought.

Much of social media is a giant meme lottery. At any given time, there's a certain finite appetite for memes. Memes work because they're shared culture. In other words, the more a meme is shared, the more valuable it becomes, and the more likely people are to share it further. There are way more videos that are "eligible" to become memes—videos that are quirky and different enough in some way—than our collective attention will allow. So, which content will win the meme lottery? It's pretty much random as far as we can tell.

In fact, if there were a science to it—say, videos with lots of emotions tend to do better—then social media content creators would quickly capitalize on it, flood the market with that type of video, and the pattern would no longer hold true. It's a lot like trying to beat the stock market. If there were a strategy to predict which stocks would go up, it would stop working as soon as people got wind of it.

Research on X (formerly Twitter) backs this up; researchers have found it essentially impossible to predict a tweet's popularity by analyzing its content using machine learning.⁴⁹ Perhaps more sophisticated methods would have helped, but based on everything we know about the dynamics of social media, it's also quite possible that popularity is simply highly random and not predictable in advance.

As further evidence that virality is not primarily based on quality, consider that videos have become viral for being so *awful* that people couldn't stop watching them, like Rebecca Black's "Friday," a song with overly simplistic lyrics about the joys of the weekend and a heavily auto-tuned performance. Interestingly, Black rode the popularity of this video to an unironically successful music career.

The unpredictability of success is well known to today's social media influencers. It is common for someone to parlay one viral video into a whole career. After "Gangnam Style" went viral, PSY became internationally successful (he was previously successful in South Korea). Charli D'Amelio, TikTok's most well-known star, similarly got her start when a few videos went viral. She is surprised by her success. "I consider myself a normal teenager that a lot of people watch, for some reason. It doesn't make sense in my head, but I'm working on understanding it."⁵⁰ But knowing what we know, this shouldn't be a surprise.

Social media companies have leaned into the unpredictability of success. The recommendation algorithms that influence people's social media diets seem to foster an even more acute rich-get-richer effect than word-of-mouth information propagation. On top of this, the design of these apps makes them addicting for creators by showing popularity counts and encouraging constant social comparison.

For the most part, all this is fine. All that's happened is that one way of identifying mass entertainment has been (partly) replaced by another. In the past, we relied on the judgment of a record producer or a TV studio. Today we rely on the wisdom of the crowd, the vagaries of the algorithm, and a large element of randomness.

There are many advantages to this lack of gatekeepers. We don't think those gatekeepers had any good way of judging

quality anyway. Bypassing them allows a larger set of creators to be potentially successful.

But there are problems. Just as content can go viral for good reasons, it can also go viral for bad ones. A woman named Justine Sacco boarded a flight from London to South Africa in 2013. Just before getting on, she posted a tweet intended to satirize Western ignorance about HIV/AIDS in Africa. When she landed, she saw that she had a text from someone she hadn't spoken to since high school: "I'm so sorry to see what's happening."

What was happening was that she had become the dreaded "main character of Twitter." Her joke was tasteless, and her satire of racism was mistaken as literal racism. People started tweeting about it out of a sense of justice, calling for her employer to fire her. As more and more people joined in, it morphed into entertainment, a kind of global digital bloodlust. The hashtag #HasJustineLandedYet started trending worldwide, with people wanting to see her shock and shame when she landed. Someone went to the Cape Town airport to take pictures. She was fired soon after. Her relationships and career were ruined, and she found it hard to find a job.⁵¹

Sacco's story is particularly awful, but it is not an isolated incident. This type of coordinated mass harassment for actual or perceived transgressions has become a regular feature of social media. Its unpredictability makes it terrifying.

Another negative effect of virality is on politics. Note that virality is not completely unpredictable. Content that is simply mediocre is unlikely to go viral, which is for the best. But some of the patterns that do exist are deeply problematic. Research shows that more partisan and more negative content gets more reach. Politicians and influencers know this well, and many of them change what they post accordingly.⁵²

Viral content is not a random sample of what other people post, much less what they think. Yet it is what we see day in and day out on social media, so we end up using it as a barometer of collective opinion. And we end up with a skewed picture. Perhaps as a result, people in the United States overestimate how polarized we actually are as a country. In other words, perceived polarization is higher than actual polarization. Could it be that this misperception in turn increases polarization, in a self-reinforcing destructive cycle?⁵³

From Individuals to Aggregates

Isaac Asimov's *Foundation* series of novels centers on the fictional science of psychohistory, which predicts future events by applying statistics and sociological knowledge to historical patterns. The key idea is that even though individual behavior isn't predictable, when we look at sufficiently large groups of people, individual randomness averages out and clear patterns emerge.

It's a tempting idea, and it's certainly intuitively plausible. After all, we can't predict individual road accidents, but we can say with virtual certainty that Los Angeles will have more accidents than Boise, Idaho, tomorrow, just based on their populations. In the business world, demand forecasting is useful, even critical.⁵⁴ Airlines would quickly go out of business if they couldn't forecast a few months out roughly how many people would want to fly between cities A and B on a given day—too many flights and they will run half empty; too few and they will lose passengers to competitors.

Quotidian business affairs are one thing, but the exciting prospect for AI is being able to predict elections, wars, pandemics, and so forth—the things that really affect our lives. And here things don't look so good.

It's not for lack of trying. One ambitious effort is the theory of cliodynamics by Peter Turchin, which applies mathematical models to populations.⁵⁵ Turchin argues that there are two-generation-long (or roughly fifty-year-long) cycles of violence and stability. His models are similar to those used by biologists who mathematically model the cycles of animal populations over generations; he applies these models to human societies. While Asimov didn't specify how psychohistory worked, cliodynamics sounds like it would fit the bill.

But it is unclear if cliodynamics really works, and it remains controversial. So do other theories, such as Peter Zeihan's prediction of the impending collapse of the current world order, including food shortages, financial instability, and political chaos.⁵⁶ In fact, predictions about geopolitical events don't have a good track record, whether using AI or human experts. Experts famously failed to predict the collapse of the USSR, despite a whole cottage industry of pundits who specialized in Cold War analysis.⁵⁷

Why is even predicting outcomes about aggregates hard in many cases? The answer becomes clear when we look at why cliodynamics, for example, has been controversial. Turchin's theory relies in part on a painstakingly compiled dataset of "political units," ranging from villages to empires, showing cyclic patterns of violence and stability. But there aren't enough historical examples to be able to rigorously show the patterns that Turchin claims to have discovered. The dataset is his crown jewel, but as of 2017, it had only 456 units.⁵⁸

Why is the dataset so small? First, the total number of political units that have ever existed is relatively small (say, compared to the number of individuals available for a project like *Fragile Families*). And the data we do have is hard to compile. Without knowing *a priori* which variables are important,

those compiling the data must describe each political unit in significant detail—using 1,500 variables. Not only is the dataset small, but it might also be a biased and unrepresentative sample of political units despite the researchers’ best efforts. The patterns that hold at a small scale, say in villages, might not hold in larger ones. Finally, the units go way back in time—to the Neolithic era—and patterns from one era might not hold in another.

So while this dataset might reveal basic, robust patterns such as the link between economic hardship and political instability, it’s unlikely that one could build complex statistical models on top of a dataset like this to tease out nuanced phenomena or generate accurate predictions about the future.

Paucity of data isn’t the only problem. To illustrate, let’s turn to disease prediction.

Flu is a seasonal disease. Every year, the Centers for Disease Control (CDC) forecast how the flu season is going to play out, which helps healthcare providers better prepare for any potential surge in patients. We have a pretty good idea of how good these forecasts are because the CDC runs an open competition called FluSight.⁵⁹

The models in the competition have consistently gotten better over the years. They are much more accurate than using simple baselines like the number of cases that occurred in a given week and region in the previous flu cycle.

But are they good enough? Recall that we think the right way to answer this question is to look at practical utility, not specific numerical thresholds. Well, people don’t check the likelihood of getting the flu before planning a party, so we haven’t yet reached a point where flu predictions are useful for everyday people. But today’s models are good enough that the CDC has continued to host FluSight year after year. And the models have already proved to be useful to healthcare providers.⁶⁰

In comparison, pandemic prediction is abysmal. We didn't see the 2020 COVID pandemic coming. A couple of months after it started, experts still had no clue how severe it would be—forecasts were all over the place. As of 2024, the figure for U.S. death tolls is around 1.2 million according to the CDC, much higher than experts initially predicted.

What are the differences between flu and COVID? First, COVID was caused by a chance event. Pandemics more generally tend to be caused by mutations, which are random events that can't be predicted. The *risk* of damage from a pandemic was extremely well known, but we couldn't have known when it would materialize and in what form.^{61,62,63}

Flu, too, mutates every year, which is why predicting it a year out is essentially impossible.* Still, seasonal flu mutations have a limited range, and each year, the mutations vary only slightly. So even if we can't forecast seasonal variation, we know roughly what's going to happen. With a new pandemic, our immune systems might be relatively defenseless to the pathogen, so the variance is a lot higher.

What about short-term forecasting after the COVID pandemic had already started? After all, even flu forecasting is only useful a few weeks out. But there are many reasons even short-term forecasting of COVID doesn't work well. While historical flu averages provide a strong baseline, COVID has no history, because it is not (or was not initially) a seasonal disease.

Another critical reason: COVID forecasts affect the outcomes being predicted. That is, after all, the point of COVID

* There are other reasons as well. For example, in 2020 there were few cases of flu due to the impact of social distancing steps taken to prevent COVID-19. As a result, the FluSight prediction competition for that year was canceled.

forecasting—so that people can adjust their behavior to avoid devastating surges. Note that this is subtly different from flu forecasting. Flu forecasting’s main goal is to help providers allocate resources when there is a surge, not to prevent the surge itself. (On the other hand, if flu forecasts ever got so good that many people started adjusting their behavior accordingly to avoid it, the outcome being predicted would be affected, making the forecast accuracy self-defeating.)

So if a surge is forecast and people modify their behavior to prevent the surge, shouldn’t that be considered a success? Not quite. The way that governments, companies, and individuals react is a bit more subtle. When COVID cases were trending up, people took protective measures. Schools shut down. Governments limited gatherings. Many people masked up. But when cases came down, the opposite happened, since the cost of social isolation is high, and people didn’t want to lose economic productivity. Masks came off, people started to return to the office and eat at restaurants again. These sorts of adjustments happened repeatedly. As a result, COVID prevalence stayed in a kind of knife-edge equilibrium for months or years in many places.

The numbers bear this out. The so-called basic reproduction number of COVID is around 3, which means that if people had gone about life as usual, each new person would have infected three new people, and the number of infections would have rapidly snowballed until most of the susceptible people in the population were infected. Getting the effective reproduction number to around 1.0—which means the number of cases is neither growing nor dwindling rapidly—required massive social changes. But having made the massive sacrifices needed to go from around 3 to 1.0, we collectively decided *not* to exert, say, 10 percent more effort, which would have brought the number

down to 0.8. That doesn't seem like a big change, but it would have meant that rather than remaining stable, the number of cases would have halved roughly every two weeks!*

In other words, COVID's reproduction number in many places around the world was remarkably stable, hovering around 1.0. Figure 3.1 shows an illustrative example. Whether and when an individual would be infected might be highly unpredictable, but as an aggregate, societies accomplished something quite fascinating. Variation between regions in COVID outcomes was driven largely by events that led to this equilibrium being disturbed and the knife falling in one direction or another. What were those events? Things like new mutant variants, vaccines becoming available, superspreader events, or a government instituting a major tightening or loosening of policies in response to social unrest. The equilibrium tended to fail when the shock introduced by the event was big enough that people's behavioral adaptations (such as working from home or returning to the office) weren't able to compensate.

That brings us to our critical point: the task of predicting whether and when these shocks might occur looks nothing like the regular business of epidemiological modeling. These are not the aggregates of millions of individual events. They are in fact individual events that have massive effects.⁶⁴

As an extreme case of the unpredictability of COVID deaths, consider China. For most of the first three years of the pandemic, the number of deaths was much smaller compared to almost any other country. But in late 2022, the Chinese people's frustration with their government's stringent "zero COVID" policy boiled over. In December of that year, Chinese President

* This calculation assumes that the "generation time" is five days, that is, the duration between someone getting infected and their infecting another person.

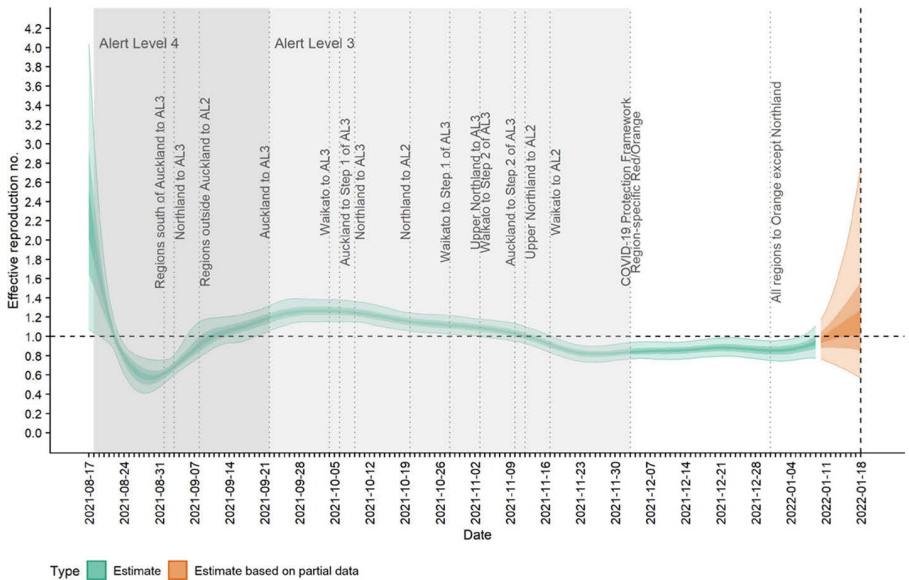


FIGURE 3.1. A case study of the effect of COVID interventions in New Zealand. The graph shows the reproduction number of COVID-19 in New Zealand over time, with ribbons showing uncertainty. The initial (“basic”) reproduction number, before interventions began, is around 3. After this quick initial spread, measures like social distancing decreased the rate of growth of the epidemic, seen in the downward curve of the reproduction number. The number then stays remarkably close to 1.0 throughout the months-long period represented in the chart. This kind of unstable equilibrium, where the virus was neither unchecked nor brought under control, was seen in many places around the world.

(Source: Binny RN, Lustig A, Hendy SC, Maclaren OJ, Ridings KM, Vattiato G, Plank MJ. “Real-Time Estimation of the Effective Reproduction Number of SARS-CoV-2 in Aotearoa New Zealand.” *PeerJ* (October 2022):e14119.)

Xi Jinping decided to end the policy, and the floodgates opened. According to many estimates, there were over a million deaths in a couple of months.⁶⁵ It had been widely predicted that this would happen—China’s zero COVID policy would have to end at some point, and the virus would then rip through the

population—but there was no good way to know how long the policies would stay in place and *when* the deaths would happen.

Recap: Reasons for Limits to Prediction

We've looked at the limits to prediction in different areas: weather, life outcomes, cultural products, criminal justice, social media, and pandemics. In some areas, there are strong limits to prediction. In others, there's steady progress toward better prediction, even if we don't know how accurate those predictions will get.

We've also seen a few broad reasons for limits to prediction. First, in many domains, there are limitations due to the data that is (or could possibly be) available. In the Fragile Families Challenge, it is possible that increasing the size and granularity of the data could lead to better predictions. It's also possible that for accurate predictions, we would need data on more humans than exist in the world (the "8 billion problem").

Another possibility is that we're not observing the features that could enable better predictions. Recall the case of the supportive neighbor supplying help and blueberries to a struggling student. There might be hundreds of such features that we can never observe about a person, because they are too specific for even the largest data collection process that we could undertake.

A different source of difficulty is the prediction task itself being too hard. For example, there could be shocks that no model could predict. In life outcome prediction, such shocks could take the form of accidents or winning the lottery. In public health, these could include a random mutation that makes a disease spread much more quickly than anticipated.

Prediction could also be hard because of feedback loops, like the amplification of small initial advantages. Such amplification

is pervasive across domains: a helpful boss in one's first job, an early review for a film or book that boosts later sales, or a retweet by a famous X/Twitter account could all lead to accumulating advantages that would have been hard to predict early on.

Finally, many of the prediction tasks we've seen involve strategic decisions, such as societal responses to COVID. When the number of COVID cases increased, additional preventive mechanisms reduced the spread. When they decreased, restrictions were eased.

In short, some existing limits to predictability could be overcome with more and better data, while others seem intrinsic. In some cases, such as for cultural products, we don't expect predictability to get much better at all. In others, such as predicting individuals' life outcomes, there could be some improvements but not drastic changes. Unfortunately, this hasn't stopped companies from selling AI for making consequential decisions about people by predicting their future. So, it is important to resist the AI snake oil that's already in wide use today rather than passively hope that predictive AI technology will get better.

In the next two chapters, we'll turn to generative AI, a technology which, unlike predictive AI, has been advancing rapidly.

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